

Basic Tools Constructing Insurable Risk Portfolios

Colombia Workshop

Edward (Jed) Frees, University of Wisconsin - Madison

17 October 2026

Outline of Presentation

- Risk Measures
- Risk Retention Functions
- Dependent Risks
- Constrained Optimization

Recall the Problem

- There are p risks, $X_j, j = 1, \dots, p$
- For each risk, the risk owner
 - retains some risk, $g_j(X_j)$
 - and cedes risk to another party $X_j - g_j(X_j)$
- The cost of ceding these risks is denoted as RTC (risk transfer cost)
 - For example, $RTC = E \left\{ \sum_{j=1}^p X_j - g_j(X_j) \right\}$
- The total retained risk is $g(\mathbf{X}) = \sum_{j=1}^p g_j(X_j)$
 - Capture this uncertainty with a risk measure, $RM[g(\mathbf{X})]$
- Our goal is to quantify this trade-off for different types of retention plans

Risk Measures

2.1 Measures of Uncertainty

Let X denote a generic risk.

- As benchmarks, we use classic measures of uncertainty
 - the variance, $\text{Var}(X) = \text{E}[X - \text{E}(X)]^2$
 - the standard deviation (positive square root of the variance)
- The value at risk is $\text{VaR}_\alpha[X] = \inf\{x : F_X(x) \geq \alpha\}$
 - We also denote this as $\text{VaR} = F_\alpha^{-1}$, the α quantile
 - To distinguish the value of risk (VaR) from the variance (Var), we follow a common practice in the literature and capitalize the **R** in the value at **R**isk.
- The expected shortfall, $\text{ES}_\alpha(X) = \text{E}(X|X > F_\alpha^{-1})$
 - the expected amount given that the loss exceeds the VaR .

Quantiles and Value at Risk

- Random variable X , distribution function F , inverse F^{-1} .
 - For a fraction $0 < q < 1$, we can write the q th quantile as $q_F = F^{-1}(q)$.
- To simulate outcomes from F , use the **inverse transform method**.
 - Start by simulating a uniform random number, U_r . The convention is the range is $[0, 1]$.
 - Next, define $X_r = F^{-1}(U_r)$.
 - Repeat $r = 1, \dots, R$ for a large number R replications.
- This method works well for *dependent variables*, as will be seen.
- For additional background on quantiles, see for example, [Loss Data Analytics, Section 4.1.1.3](#).
- For additional background on simulating random variables, see for example, [Loss Data Analytics, Section 6.1](#)

Quantile-based Risk Measures

- VaR and ES are types of quantile-based risk measures
- For comparison purposes, the book also employs two other quantile-based risk measures
 - The *GlueVaR*, defined as

$$GlueVaR = \omega_1 VaR_\alpha + \omega_2 ES_\alpha + \omega_3 ES_{\alpha+\beta}.$$

- The **range value at risk** is defined as

$$RVaR_{(\alpha,\beta)}(X) = \begin{cases} \frac{1}{\beta} \int_{\alpha}^{\alpha+\beta} VaR_a(X) da & \text{if } \beta > 0 \\ VaR_\alpha(X) & \text{if } \beta = 0 \end{cases}.$$

Risk Retention Functions

- Think about p risks (such as 15), denoted as $X_1, \dots, X_p$
- Each risk is subject to an insurance contract that specifies:
 - $g_j(X_j; \theta)$ - the amount retained
 - $X_j - g_j(X_j; \theta)$ - the amount transferred to another party

These contracts can be complex, so we start at the foundations:

- Suppose that $g(x; \theta = c) = c \cdot x$, a *coinsurance* or *quota-share*
 - For example, your insurer pays 80% of your prescription drugs expenses, you pay 20%
- Suppose that $g(x; \theta = u) = \min(x, u)$, known as *deductible* or *excess of loss*
 - For example, you retain claims from your auto accident up to 250, the insurer pays the excess
- I refer to c, u, θ as **risk retention** parameters

2.2.1 A Risk Retention Function

An important special case that covers many situations of interest is

$$g(x; d, c, u) = \begin{cases} 0 & \text{if } x < d \\ c(x - d) & \text{if } d \leq x < u \\ c(u - d) & \text{if } x \geq u \end{cases} .$$

Here, d is the deductible, c is the coinsurance amount, and u is the upper limit of coverage. This function can also be expressed as

$$g(x; d, c, u) = c(x \wedge u - x \wedge d) .$$

The wedge symbol $\wedge$ is used for the minimum operator and is defined as $x \wedge z = \min(x, z)$.

2.2.3 Risk Measures

Quantiles preserve location and scale transformations. For example,

$$VaR_{\alpha}(a_1X - a_0) = a_1 \times VaR_{\alpha}(X) - a_0, \quad \text{for } a_1 > 0.$$

This holds for all quantile-based measures, including *ES*, *RVaR*, and *GlueVaR*.

For an upper limit only policy ($d = 0, c = 1$), one can check that

$$F_{g(X;u)}(z) = \Pr(\min(X, u) \leq z) = \begin{cases} F(z) & \text{if } z < u \\ 1 & \text{if } z \geq u \end{cases}.$$

So

$$VaR_{\alpha}[g(X; u)] = \begin{cases} F_{\alpha}^{-1} & \text{if } \alpha < F(u-) \\ u & \text{if } \alpha \geq F(u-) \end{cases}.$$

Using the figure for the distribution of $F_{g(X;\theta)}(z)$, check that

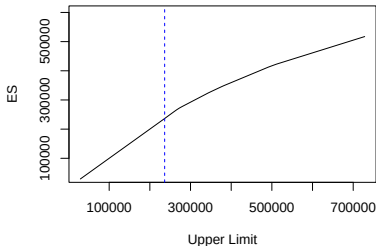
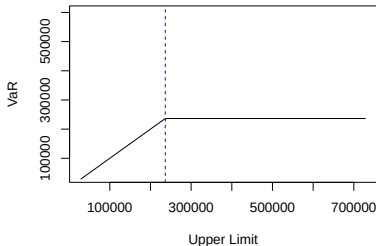
$$VaR_{\alpha}[g(X)] = \begin{cases} 0 & \text{if } \alpha < F(d) \\ c(F_{\alpha}^{-1} - d) & \text{if } F(d) \leq \alpha < F(u-) \\ c(u - d) & \text{if } \alpha \geq F(u-) \end{cases} .$$

In the same way, check that

$$ES_{\alpha}[g(X)] = \begin{cases} \frac{c}{1-\alpha} \{E(X \wedge u) - E(X \wedge d)\} & \text{if } \alpha < F(d) \\ \frac{c}{1-\alpha} \{E(X \wedge u) - E(X \wedge F_{\alpha}^{-1}) \\ \quad + (1-\alpha)(F_{\alpha}^{-1} - d)\} & \text{if } F(d) < \alpha < F(u) \\ c(u - d) & \text{if } F(u) < \alpha \end{cases} .$$

Example 2.1. Property Fund - Continued.

- Assume no deductible ($d = 0$) and no coinsurance ($c = 1$).
- A confidence level of $\alpha = 0.99$.
- $\widehat{VaR}_{0.99} = 236,427$, marked with blue dashed vertical lines
- The left-hand panel of this figure shows a change in the slope for the VaR whereas the right-hand panel shows that the ES is smoother at this point.



General Risk Retention Functions

- Through simulation strategies, we will have the ability to analyze complex risk transfer agreements.
 - Recall that $g(\mathbf{X}; \theta)$ represents a risk owner's retained risk after a risk transfer agreement.
- In a one-day workshop, we will focus on basic functions such as quota share and excess of loss.
- A summary of general risk retention functions is Section 7.5 of the book.

We will go through a quick run through now.

Separate Contracts for each Risk

- Could postulate a different contract for each risk.
 - The amount retained for the j th risk is $g_j(X_j; \theta_j)$.
 - With an additive structure among retained risks

$$g(\mathbf{X}; \theta) = \sum_{j=1}^p g_j(X_j; \theta_j).$$

- Both quota share and excess of loss are special cases.
 - Could also have some risks follow quota share, others excess of loss.
 - Could use $\theta = (d, c, u)$ where d is the deductible, c is the coinsurance amount, and u is the upper limit of retention.
- Working with additive structures will simplify the numerical convergence of our optimization procedures.

Combining Risks

Combined Deductibles from Bundled Insurance Contracts

- In personal insurance, combined deductibles are common when insurers “bundle” together several types of insurance.
- For example, in travel insurance X_1 might represent the loss due to luggage damage and X_2 is the loss due to personal injury.
- Then, a type of insurance policy covers

$$Y_{insurer} = [(X_1 - d_1)_+ + (X_2 - d_2)_+ - d_{combined}]_+.$$

- d_1 is the deductible for luggage claims and $(X_1 - d_1)_+$ is the luggage excess.
- d_2 is the deductible for personal injury and $(X_2 - d_2)_+$ is the personal injury excess.
- The insurer is responsible for the excess over a combined deductible, $d_{combined}$, over these two separate excesses.

Example. Global Protections Reinsurance Coverage.

- Consider an agreement where the reinsurer's portion of the total risk is

$$Y_{reinsurer} = \min \left(\sum_{j=1}^p \min[(X_j - u_{1,j})_+, u_{2,j}], AAL \right).$$

- $u_{1,j}$ represents a reinsurer's deductible,
- $u_{2,j}$ represents is upper limit for the j th risk, and
- AAL stands for the annual aggregate limit AAL .

Example. Separate Limits by Peril.

- A cause of loss is known as *peril*
- It is common for there to exist special arrangements such as sublimits made for each peril.
- For example, one might introduce a sublimit u_1 that limits how much of risk X_1 can be transferred.
 - Then, if X_2 represents all other risks and $u_{overall}$ is an overall limit, we can use

$$Y_{firm} = (X_1 \wedge u_1 + X_2) \wedge u_{overall}$$

to express the amount retained by a firm.

Frequency Severity Approaches to Modeling Risks

- For risks that occur frequently, it is convenient to define risk transfers on a *per event* or *per occurrence of a claim* basis.
- the *collective risk* model - decompose the overall type j loss amount into
 - a number of claims in a year, say N_j , and
 - examine each individual claim outcome, $X_{j1}, \dots, X_{jN_j}$.
 - we express aggregate type j losses as $X_j = \sum_{i=1}^{N_j} X_{ji}$.
- Especially for high frequency risk types such as auto insurance, it is common to define risk transfer mechanisms (such as deductibles and upper limits) for each claim.
 - For example, a firm might retain claims up to u_j and so the total retained claims is

$$Y_{j,retained} = \sum_{i=1}^{N_j} X_{ji} \wedge u_j.$$

Frequency Severity Approaches to Modeling Risks 2

- For some coverages such as property insurance, there may be many claims due to a common event such as earthquakes or hurricanes.
 - The amount transferred may be of the form

$$Y_{j,transfer} = \left[\sum_{i \in \text{event}} (X_{ji} - d_j)_+ \right] \wedge u_j.$$

- For each claim, $(X_{ji} - d_j)_+$ represents the amount in excess of a deductible d_j .
- The sum within the square braces is the total excess for all claims but the transferred amount is limited u_j .

Dependent Risks

- In the insurance sector, risks are almost never independent.
- There are many ways to model dependence, latent variables being a common approach
- Instead, we will utilize **copulas**.
 - Still, much of the material in this course works well if you want to use another approach for modeling dependence.
- **Definition:** A *copula* is a multivariate distribution function with uniform marginals.
- Copulas enable us to work with multiple risks having different distributions,
 - perhaps the first one being gamma distributed, the second with a Pareto distribution, the third having a normal distribution.
 - Even though each marginal distribution may differ, we are able to link (or *couple*) them using a copula.

4.2.1 Copulas Defined

- Specifically, let $\{U_1, \dots, U_p\}$ be p uniform random variables on $(0, 1)$. (**Not** necessarily independent!)
 - Their distribution function is a **copula**

$$C(u_1, \dots, u_p) = \Pr(U_1 \leq u_1, \dots, U_p \leq u_p).$$

- To extend this, consider arbitrary *marginal* distribution functions $F_1(\cdot), \dots, F_p(\cdot)$.
 - Define a multivariate distribution function F using the copula such that

$$F(y_1, \dots, y_p) = C(F_1(y_1), \dots, F_p(y_p)).$$

- For any $F(\cdot)$, there *always* exists a copula (Sklar's Theorem).

Two Examples of Copulas

Frank's copula, for $p = 2$ variables

$$C_F(u_1, u_2) = \frac{1}{\gamma} \log \left(1 + \frac{(\exp[\gamma u_1] - 1)[\exp(\gamma u_2) - 1]}{\exp(\gamma) - 1} \right).$$

- This is a bivariate distribution function.
 - Check that $C(u_1, 1) = u_1 \implies$ marginals have a uniform (on $[0, 1]$) distribution $\implies C$ is a copula.
 - γ is a dependence parameter.
 - Simple, but only useful for $p=2$ variables.

- The *normal*, or *Gaussian*, copula. In $p = 2$ dimensions, this is

$$C_N(u_1, u_2) = \Phi_2 \left(\Phi^{-1}(u_1), \Phi^{-1}(u_2) \right).$$

- The symbol $\Phi(\cdot)$ represents the cumulative standard normal *df* and $\Phi^{-1}(\cdot)$ is the corresponding quantile function
 - In R these can be evaluated using the functions `pnorm` and `qnorm`, respectively.
 - The symbol $\Phi_2(\cdot)$ represents the a bivariate normal *df* that, in R, can be evaluated using the function `mvtnorm`.
 - The normal is also readily available in multiple dimensions.
- We use the normal copula extensively. But, for almost every application, other copulas readily apply.
 - The **extent** of the dependence matters, more so than the shape of the copula.

4.2.2 Copulas, Risk Retention, and Simulation

Copulas are useful for risk retention problems, especially when combined with simulation techniques.

Example 4.2.1. Excess of Loss with Two Risks

- Let X_1 be gamma and X_2 be Pareto distributed.
- Use a normal copula with dependence parameter $\rho_S = 0.30$.
- The risk owner retains $S(u_1, u_2) = X_1 \wedge u_1 + X_2 \wedge u_2$.
- The distribution and moments are hard to compute analytically, easy with simulation

Simulation Approach

- Using simulation, create a random realization of the bivariate uniform random variates (U_{1r}, U_{2r}) . In the R package `copula`, we do this with the function `rCopula`.
- Use the inverse of the distribution function to create random realizations of the random variables, $X_{1r} = F_1^{-1}(U_{1r})$ and $X_{2r} = F_2^{-1}(U_{2r})$.
 - For F_1^{-1} , in R we use the function `qgamma`.
 - For F_2^{-1} , in R we use the function `qpareto`.
- Create a random realization of the limited sum,
 $S_r(u_1, u_2) = X_{1r} \wedge u_1 + X_{2r} \wedge u_2$.
- Repeat this for a large number, R , of replications.

Example 4.6 Illustrative Code

```
# Packages
library("actuar") ; library("copula")
# Set Simulation Parameters
nsim <- 20000
set.seed(202020)
# Marginal distribution parameters
risk1shape <- 1 ; risk1scale <- 10
risk2shape <- 1 ; risk2scale <- 10
# Normal, or Gaussian, Copula
norm.cop <- copula::normalCopula(param=0.3, dim = 2,'dispstr' ="un")
UCop <- copula::rCopula(nsim, norm.cop)
# Marginal Distributions
X1 <- qgamma(p=UCop[,1],shape = risk1shape, scale = risk1scale)
X2 <- actuar::qpareto(p=(UCop[,2]),shape = risk2shape,
                      scale = risk2scale)

# Risk Retention Function
g0 <- function(u1,u2){
  S <- pmin(X1,u1) + pmin(X2,u2)
  return( S )
}
```

Constrained Optimization

Algorithmic Framework - Risk Retention

- Define the retained risk

$$g(\mathbf{X}; \theta) = \sum_{j=1}^p g_j(X_j; \theta)$$

- Denote the uncertainty as $RM[g(\mathbf{X}; \theta)]$.
 - For the RM (risk measure), variance is a possibility. I typically use expected shortfall.
- The risk transferred is $X_1 + \dots + X_p - g(\mathbf{X}; \theta)$.
 - The cost of the transfer is denoted as $RTC(\theta)$
 - It is evaluated using an insurance pricing principle, starting with an expectation (augmented for administrative costs)

Risk Retention Optimization Problem

We seek to find those risk retention parameters in the vector θ that

minimize θ	$RM[g(\mathbf{X}; \theta)]$	<i>uncertainty</i>
subject to	$RTC(\theta) \leq RTC_{max}$	<i>budget restriction</i>
	$\theta_j \geq 0$	<i>parameter restrictions</i>

- Minimize the uncertainty in retained risks summarized by $RM[g(\mathbf{X}; \theta)]$
- Limited to the aptly termed “budget” constraint $RTC(\theta) \leq RTC_{max}$.
 - The values of θ that satisfy this constraint are known as the *feasible set*.
 - Could easily add other parameter restrictions (e.g., $\theta \leq 1$ for coinsurance)
- Behind the scene, computation is complex
 - Not a convex problem (like machine learning)
 - Requires simulation techniques to compute

3.3.2 An Upper Limit Contract

One risk, one retention parameter.

Consider an upper limit contract with fair costs, so the risk transfer cost is

$$RTC(u) = E(X) - E(X \wedge u) = \int_u^{\infty} (1 - F(z)) dz.$$

With this expression, we see that $RTC(u)$ is monotonic in u . Thus $RTC(u) \leq RTC_{max} \iff u \geq u_{max}$. Using the value at risk for the risk measure, our problem is

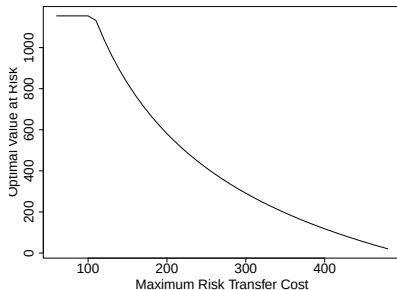
$$\begin{array}{ll} \text{minimize}_u & \min(F_{\alpha}^{-1}, u) \\ \text{subject to} & u \geq u_{max}. \end{array}$$

The risk measure is also monotonic in u , so you can check that

$$VaR_{\alpha}(X \wedge u)_{opt} = VaR_{\alpha}(X \wedge u_{max}) = \begin{cases} u_{max} & \text{if } u_{max} \leq F_{\alpha}^{-1} \\ F_{\alpha}^{-1} & \text{if } u_{max} > F_{\alpha}^{-1} \end{cases}.$$

Example. Pareto Distribution.

- The quantile is $F_{0.90}^{-1} = 1154.435$.
- First use RTC_{max} to determine u_{max} (not shown)
- Then determine $VaR_{\alpha}(X \wedge u)_{opt}$ (shown)
- Note that the value at risk is bounded by $F_{0.90}^{-1}$.



3.3.3 The Lagrangian

- The Lagrange method - minimize (or maximize) a function subject to equality constraints.
- Consider only those values of θ so that $RTC(\theta^*) = RTC_{max}$.
- Then, the corresponding Lagrangian is

$$LA(\theta, LM) = RM[g(X; \theta)] + LM \{RTC(\theta) - RTC_{max}\}.$$

- One determines the parameters in θ and the new parameter LM that optimize the function LA .
 - Turns a constrained problem into an unconstrained problem.
- To interpret the new parameter LM , take derivatives.

$$LM = - \frac{\partial_{\theta_i} RM[g(X; \theta)]}{\partial_{\theta_i} RTC(\theta)} = RM_i^2.$$

- This is a ratio of the differential changes (Section 2.3).
 - Called the **risk measure relative marginal** change, or RM^2 change, for short.

3.3.4 A General Constrained Optimization Problem

$$\begin{array}{ll} \text{minimize}_{\theta} & f_0(\theta), \quad \theta \in R^p \\ \text{subject to} & f_{in,j}(\theta) \leq 0 \quad j = 1, \dots, m_{in} \\ & f_{eq,j}(\theta) = 0 \quad j = 1, \dots, m_{eq}. \end{array}$$

- In many instances, we are able to solve a constrained optimization problem numerically by relying on software.
- Extend the Lagrangian to include inequality constraints

$$LA(\theta, \mathbf{LMI}, \mathbf{LME}) = f_0(\theta) + \sum_{j=1}^{m_{in}} LMI_j f_{in,j}(\theta) + \sum_{j=1}^{m_{eq}} LME_j f_{eq,j}(\theta),$$

- $\mathbf{LMI} = (LMI_1, \dots, LMI_{m_{in}})'$ - Lagrange multiplier vector for inequalities
- $\mathbf{LME} = (LME_1, \dots, LME_{m_{eq}})'$ - Lagrange multiplier vector for equalities.

Code for Several Optimization Runs - A Frontier!

```
meanSum = mean( g0_XL( u1=Inf, u2 = Inf) )
# Objective function to minimize expected shortfall
eval_f <- function(par, ...) {
  u1 <- par[1]
  u2 <- par[2]
  SumStats.fct( g0_XL( u1, u2 ) )[4]
}
# Inequality constraints (hin >= 0)
eval_g <- function(par, ...) {
  u1 <- par[1]
  u2 <- par[2]
  # Constraint: RTCmax - (meanSum_retained) >= 0
  RTCmax - (meanSum - SumStats.fct( g0_XL( u1, u2 ) )[1])
}
# Initial values for u.lim_alabama
u_init <- c(2, 1)

SumResults <- matrix(0, 9, 7)
# Loop through different RTCmax values
for (i_val in (1:9)) {
  RTCmax = (i_val/10) * meanSum
  # Run the optimization
  result_alabama <- auglag(
    par = u_init,
    fn = eval_f,
    hin = eval_g,
    lower = c(0, 0), # Add non-negativity constraint for u1 and u2
    control.outer=list(method="nlnmb",trace=FALSE) )
  # Extract the optimal portfolio proportions
  u.lim_alabama <- result_alabama$par
  SumResults[i_val,] <- c(RTCmax, SumStats.fct( g0_XL(u.lim_alabama[1],u.lim_alabama[2] ) ), u.lim_alabama
}
```